SPRING MID SEMESTER EXAMINATION-2025

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Database Management System
CS20006/ IT20006/ CC20006/ CM20006/ CS2004

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]
   a. Using Armstrong's axioms, prove that if A→B and BC→D then AC→D.
   b. We can convert weak entity sets into strong entity sets by simply adding the appropriate attributes. Then, why are weak entity sets needed?
   c. What are recursive relationships? Give one example where this is required.
   d. Explain the difference between two-tier and three-tier application architectures. Which is better suited for web applications?
   e. Consider a relation R with attributes $\{a_1, a_2, a_3, a_4, a_5\}$ and a_4 as the primary key. Determine the total number of super keys for relation R.

2. An educational institute database needs to store information about faculty members (identified by faculty-id, with faculty-name, doj, and specialization as attributes); departments (identified by dept-id, with dept-name as attributes); projects (identified by proj-id, with proj-name, proj-location as attributes) and children of faculty members (with child-name and child-age as attributes). A department can have many faculty members and faculty member can teach in more than one department. Faculty members can work on different projects.

   A department can have many projects and a project can belong to at most one department. Each department is managed by one HOD, who is a faculty member. A child must be identified uniquely by name when the parent (who is a faculty member; assume that only one parent works for institute) is known. We are not interested in information about a child one the parent leaves the institute.

   Answer the following questions

   i) Draw the ER diagram that captures the above information.

   ii) Translate the ER diagram into relations. Also identify the primary key and foreign keys. [3+2 Marks]

3. Explain the Database System Architecture and its significance of each level, support your explanation with a diagram. [5 Marks]

4. Consider the following relational database schema consisting of the four relation schemas:

   passenger (pid, pname, pgender, pcity)

   agency (aid, aname, acity)

   flight (fid, fdate, time, src, dest)

   booking (pid, aid, fid, fdate)

   Answer the following questions using relational algebra queries or SQL;

   a. Find the agency names for agencies that located in the same city as passenger with passenger id 123.
   b. Get the details about all flights from Chennai to New Delhi.
   c. Find only the flight numbers for passenger with pid 123 for flights to Chennai before 06/11/2020.
   d. Find the passenger names for passengers who have bookings on at least one flight.
   e. Find the passenger names for those who do not have any bookings in any flights.

   [1 Marks X 5]

5. Consider two entity sets A and B that both have the attribute X (among others whose names are not relevant to this question).
   a. If the two X's are completely unrelated. How should the design be improved?
   b. If the two X's represent the same property and it is one that applies both to A and to B. How should the design be improved? Consider three sub-cases.
      i. X is the primary key for A but not B
      ii. X is the primary key for both A and B
      iii. X is not the primary key for A nor for B

   [2+3 Marks]

*** Best of Luck ***